

JEE Mains 2019 Chapter wise Question Bank

Permutation and Combination – Questions

Q1

Consider a class of 5 girls and 7 boys. The number of different teams consisting of 2 girls and 3 boys that can be formed from this class, if there are two specific boys A and B, who refuse to be the members of the same team, is:

- (1) 500 (2) 200
(3) 300 (4) 350

9 Jan Morning

Q2

The number of natural numbers less than 7,000 which can be formed by using the digits 0,1,3,7,9 (repetition of digits allowed) is equal to:

- (1) 374 (2) 372
(3) 375 (4) 250

9 Jan Evening

Q3

In a class of 140 students numbered 1 to 140, all even numbered students opted Mathematics course, those whose number is divisible by 3 opted Physics course and those whose number is divisible by 5 opted Chemistry course. Then the number of students who did not opt for any of the three courses is:

- (1) 102 (2) 42 (3) 1 (4) 38

10 Jan Morning

Q4

The sum of all two digit positive numbers which when divided by 7 yield 2 or 5 as remainder is:

- (1) 1256 (2) 1465
(3) 1365 (4) 1356

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Q5

The number of functions f from $\{1, 2, 3, \dots, 20\}$ onto $\{1, 2, 3, \dots, 20\}$ such that $f(k)$ is a multiple of 3, whenever k is a multiple of 4 is :

- (1) $6^5 \times (15)!$ (2) $5! \times 6!$
(3) $(15)! \times 6!$ (4) $5^6 \times 15$

11 Jan Evening

Q6

Consider three boxes, each containing 10 balls labelled 1, 2, ..., 10. Suppose one ball is randomly drawn from each of the boxes. Denote by n_i , the label of the ball drawn from the i^{th} box, ($i = 1, 2, 3$). Then, the number of ways in which the balls can be chosen such that $n_1 < n_2 < n_3$ is :

- (1) 120 (2) 82
(3) 240 (4) 164

12 Jan Morning

Q7

Let $S = \{1, 2, 3, \dots, 100\}$. The number of non-empty subsets A of S such that the product of elements in A is even is :

- (1) $2^{100} - 1$ (2) $2^{50} (2^{50} - 1)$
(3) $2^{50} - 1$ (4) $2^{50} + 1$

12 Jan Morning

Q8

All possible numbers are formed using the digits 1, 1, 2, 2, 2, 2, 3, 4, 4 taken all at a time. The number of such numbers in which the odd digits occupy even places is :

- (1) 180 (2) 175
(3) 160 (4) 162

8 April Morning

Q9

Permutation and Combination

The sum of all natural numbers ' n ' such that $100 < n < 200$ and H.C.F. $(91, n) > 1$ is :

- (1) 3203 (2) 3303 (3) 3221 (4) 3121

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Q10

The number of four-digit numbers strictly greater than 4321 that can be formed using the digits 0, 1, 2, 3, 4, 5 (repetition of digits is allowed) is:

- (1) 288 (2) 360 (3) 306 (4) 310

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Q11

A committee of 11 members is to be formed from 8 males and 5 females. If m is the number of ways the committee is formed with at least 6 males and n is the number of ways the committee is formed with at least 3 females, then:

- (1) $m + n = 68$ (2) $m = n = 78$
(3) $n = m - 8$ (4) $m = n = 68$

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Q12

Some identical balls are arranged in rows to form an equilateral triangle. The first row consists of one ball, the second row consists of two balls and so on. If 99 more identical balls are added to the total number of balls used in forming the equilateral triangle, then all these balls can be arranged in a square whose each side contains exactly 2 balls less than the number of balls each side of the triangle contains. Then the number of balls used to form the equilateral triangle is:

- (1) 157 (2) 262 (3) 225 (4) 190

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Q13

The number of 6 digit numbers that can be formed using the digits 0, 1, 2, 5, 7 and 9 which are divisible by 11 and no digit is repeated, is:

- (1) 72 (2) 60 (3) 48 (4) 36

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Q14

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Suppose that 20 pillars of the same height have been erected along the boundary of a circular stadium. If the top of each pillar has been connected by beams with the top of all its non-adjacent pillars, then the total number of beams is :

- (1) 170 (2) 180 (3) 210 (4) 190

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Q15

The number of ways of choosing 10 objects out of 31 objects of which 10 are identical and the remaining 21 are distinct is :

- (1) $2^{20} - 1$ (2) 2^{21} (3) 2^{20} (4) $2^{20} + 1$

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Q16

A group of students comprises of 5 boys and n girls. If the number of ways, in which a team of 3 students can randomly be selected from this group such that there is at least one boy and at least one girl in each team, is 1750, then n is equal to :

- (1) 28 (2) 27 (3) 25 (4) 24

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Permutation and Combination - Solutions

Q1

(3) Since, the number of ways to select 2 girls is 5C_2 .
Now, 3 boys can be selected in 3 ways.

(a) Selection of A and selection of any 2 other boys (except B) in 5C_2 ways

(b) Selection of B and selection of any 2 two other boys (except A) in 5C_2 ways

(c) Selection of 3 boys (except A and B) in 5C_3 ways

Hence, required number of different teams

$$= {}^5C_2 ({}^5C_2 + {}^5C_2 + {}^5C_3) = 300$$

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Q2

(1) Number of numbers with one digit = $4 = 4$

Number of numbers with two digits = $4 \times 5 = 20$

Number of numbers with three digits = $4 \times 5 \times 5$
= 100

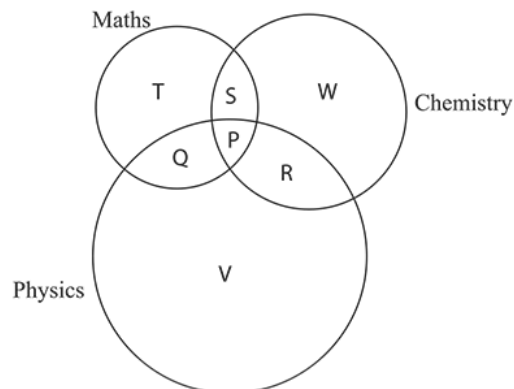
Number of numbers with four digits = $2 \times 5 \times 5 \times 5$
= 250

$\therefore$ Total number of numbers = $4 + 20 + 100 + 250$
= 374

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Q3

(4)



$$P = \{30, 60, 90, 120\}$$

$$\Rightarrow n(P) = 4$$

$$Q = \{6n: n \in N, 1 \leq n \leq 23\} - P$$

$$\Rightarrow n(Q) = 19$$

$$R = \{15n: n \in N, 1 \leq n \leq 9\} - P$$

$$\Rightarrow n(R) = 5$$

$$S = \{10n: n \in N, 1 \leq n \leq 14\} - P$$

$$\Rightarrow n(S) = 10$$

$$n(T) = 70 - n(P) - n(Q) - n(S)$$

$$= 70 - 33 = 37$$

$$n(V) = 46 - n(P) - n(Q) - n(R)$$

$$= 46 - 28 = 18$$

$$n(W) = 28 - n(P) - n(R) - n(S)$$

$$= 28 - 19 = 9$$

$\Rightarrow$ Number of required students

$$= 140 - (4 + 19 + 5 + 10 + 37 + 18 + 9)$$

$$= 140 - 102 = 38$$

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Permutation and Combination

Q4

- (4) Two digit positive numbers which when divided by 7 yield 2 as remainder are 12 terms i.e, 16, 23, 30, ..., 93

Two digit positive numbers which when divided by 7 yield 5 as remainder are 13 terms i.e, 12, 19, 26, ..., 96

By using AP sum of 16, 23, ..., 93, we get

$$S_1 = 16 + 23 + 30 + \dots + 93 = 654$$

By using AP sum of 12, 19, 26, ..., 96, we get

$$S_2 = 12 + 19 + 26 + \dots + 96 = 702$$

$$\text{required Sum} = S_1 + S_2 = 654 + 702$$

$$= 1356$$

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Q5

- (3) Domain and codomain = $\{1, 2, 3, \dots, 20\}$.

There are five multiple of 4 as 4, 8, 12, 16 and 20.

and there are 6 multiple of 3 as 3, 6, 9, 12, 15, 18.

Since, when ever k is multiple of 4 then $f(k)$ is multiple of 3 then total number of arrangement

$$= {}^6C_5 \times 5! = 6!$$

Remaining 15 elements can be arranged in $15!$ ways.

Since, for every input, there is an output

$\Rightarrow$ function $f(k)$ in onto

$$\therefore \text{Total number of arrangement} = 15! \cdot 6!$$

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Q6

- (1) Collecting different labels of balls drawn = $10 \times 9 \times 8$

$\therefore$ arrangement is not required.

$\therefore$ the number of ways in which the balls can be chosen is,

$$\frac{10 \times 9 \times 8}{3!} = 120$$

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Q7

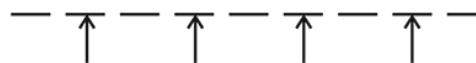
- (2) $\therefore$ Product of two even number is always even and product of two odd numbers is always odd.

$$\begin{aligned} \therefore \text{Number of required subsets} &= \text{Total number of subsets} \\ &\quad - \text{Total number of subsets having only odd numbers} \\ &= 2^{100} - 2^{50} \\ &= 2^{50}(2^{50} - 1) \end{aligned}$$

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Q8

- (1) $\therefore$ There are total 9 digits and out of which only 3 digits are odd.



$\therefore$ Number of ways to arrange odd digits first

$$= {}^4C_3 \cdot \frac{3!}{2!}$$

Hence, total number of 9 digit numbers

$$= \left({}^4C_3 \cdot \frac{3!}{2!} \right) \cdot \frac{6!}{2!4!} = 180$$

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Q9

- (4) $\therefore 91 = 13 \times 7$

Then, the required numbers are either divisible by 7 or 13.

$$\begin{aligned} \therefore \text{Sum of such numbers} &= \text{Sum of no. divisible by 7} + \text{sum of the no. divisible by 13} - \text{Sum of the numbers divisible by 91} \\ &= (105 + 112 + \dots + 196) + (104 + 117 + \dots + 195) - 182 \\ &= 2107 + 1196 - 182 \\ &= 3121 \end{aligned}$$

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Q10

Permutation and Combination

(4) 0, 1, 2, 3, 4, 5

Number of four-digit number starting with 5 is,

$$\begin{array}{|c|c|c|c|} \hline 5 & & & \\ \hline \end{array} = 6 \times 6 \times 6 = 216$$

↓ ↓ ↓
6 6 6

Number of four-digit numbers starting with 45 is,

$$\begin{array}{|c|c|c|c|} \hline 4 & 5 & & \\ \hline \end{array} = 6 \times 6 = 36$$

↓ ↓
6 6

Number of four-digit numbers starting with 44 is,

$$\begin{array}{|c|c|c|c|} \hline 4 & 4 & & \\ \hline \end{array} = 6 \times 6 = 36$$

↓ ↓
6 6

Number of four-digit numbers starting with 43 and greater than 4321 is,

$$\begin{array}{|c|c|c|c|} \hline 4 & 3 & & \\ \hline \end{array} = 3 \times 6 = 18$$

↓ ↓
(5/4/3) 3 6

Number of four-digit numbers starting with 432 and greater than 4321 is,

$$\begin{array}{|c|c|c|c|} \hline 4 & 3 & 2 & \\ \hline \end{array} = 4$$

↓
4

Hence, required numbers = $216 + 36 + 36 + 18 + 4 = 310$.

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Q11

- (2) Since, m = number of ways the committee is formed with at least 6 males
 $= {}^8C_6 \cdot {}^5C_5 + {}^8C_7 \cdot {}^5C_4 + {}^8C_8 \cdot {}^5C_3 = 78$
 and n = number of ways the committee is formed with at least 3 females
 $= {}^5C_3 \cdot {}^8C_8 + {}^5C_4 \cdot {}^8C_7 + {}^5C_5 \cdot {}^8C_6 = 78$
 Hence $m = n = 78$

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Q12

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(4) Number of balls used in equilateral triangle = $\frac{n(n+1)}{2}$

$\because$ side of equilateral triangle has n -balls

$\therefore$ no. of balls in each side of square is $= (n-2)$

According to the question,

$$\frac{n(n+1)}{2} + 99 = (n-2)^2$$

$$\Rightarrow n^2 + n + 198 = 2n^2 - 8n + 8$$

$$\Rightarrow n^2 - 9n - 190 = 0 \Rightarrow (n-19)(n+10) = 0$$

$$\Rightarrow n = 19$$

Number of balls used to form triangle

$$= \frac{n(n+1)}{2} = \frac{19 \times 20}{2} = 190$$

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Q13

(2) Given digit 0, 1, 2, 5, 7, 9

$$\begin{array}{|c|c|c|c|c|c|} \hline a_1 & a_2 & a_3 & a_4 & a_5 & a_6 \\ \hline \end{array}$$

$$(a_1 + a_3 + a_5) - (a_2 + a_4 + a_6) = 11K$$

Therefore, (1, 2, 9) (0, 5, 7)

Number of ways to arranging them

$$= 3! \times 3! + 3! \times 2 \times 2 = 6 \times 6 + 6 \times 4 = 6 \times 10 = 60$$

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Q14

(1) Total number of beams = ${}^{20}C_2 - 20 = 190 - 20 = 170$

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Q15

(3) Number of ways of selecting 10 objects

$$= (10I, 0D) \text{ or } (9I, 1D) \text{ or } (8I, 1D) \text{ or } \dots (0I, 10D)$$

Here, D signifies distinct object and I indicates identical object

$$= 1 + {}^{21}C_1 + {}^{21}C_2 + \dots + {}^{21}C_{10} = \frac{2^{21}}{2} = 2^{20}$$

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Q16

- (3) Number of ways of selecting three persons such that there is atleast one boy and atleast one girl in the selected persons

$$= {}^{n+5}C_3 - {}^nC_3 - {}^5C_3 = 1750$$

$$\Rightarrow \frac{(n+5)!}{3!(n+2)!} - \frac{n!}{3!(n-3)!} - \frac{5!}{3!2!} = 1750$$

$$\Rightarrow \frac{(n+5)(n+4)(n+3)}{6} - \frac{n(n-1)(n-2)}{6} = 1760$$

$$\Rightarrow n^2 + 3n - 700 = 0 \Rightarrow n = 25 \quad [n = -28 \text{ rejected}]$$

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